



The NSF Institute for
Artificial Intelligence & Fundamental Interactions

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Submitted by email to ostp-ai-rfi@nitrd.gov

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan

A revolution is brewing in AI+Science, and the NSF Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) appreciates this opportunity to provide insights we have gained through almost 5 years of conducting interdisciplinary research in AI and Physics to help shape policy that will strengthen the dominance of the United States in AI. We feel it is crucial for academics to contribute to discussions of policy and national strategy given the vast experience of researchers in the academic sector working across disciplines at the forefront of both scientific discovery and new technologies. Therefore, we present information to support what we believe is most essential for the future of AI in the United States: (1) **investing** in interdisciplinary research that prioritizes **robust and interpretable AI**, (2) **educating and training** an AI-enabled STEM workforce, and (3) providing **scalable computing resources** to support AI innovation.

About IAIFI

As one of the inaugural NSF AI Institutes, IAIFI serves as a nexus point for collaborative efforts to advance both Physics and AI.¹ By combining state-of-the-art research with early career talent and a growing AI+Physics community, IAIFI is enabling researchers to develop AI technologies to tackle some of the most challenging problems in physics, and transfer these technologies to the broader AI community. Additionally, IAIFI researchers are **applying physics principles to develop more robust and interpretable AI technologies**. IAIFI researchers are fusing the deep learning revolution with the time-tested strategies of deep thinking in physics to gain a deeper understanding of our universe and of the principles underlying intelligence itself. IAIFI research spans a range of disciplines, covering topics

¹<https://iaifi.org>

in Foundational AI, Physics Theory, Physics Experiment, and Astrophysics. IAIFI has demonstrated that many problems across these domains have shared solutions, since they involve similar prior knowledge, are based on the same underlying physics principles, and face common experimental and theoretical challenges. Thus, enabling researchers across AI+Physics to work together has been a transformative tool for scientific advancement.

Invest in interdisciplinary research to advance AI+Physics and prioritize robust and interpretable AI

Given the fundamental principles underlying the physical sciences, the Physics domain is especially well positioned to enhance our understanding of how AI works, as well as push the field toward robust techniques with its complex and extensive datasets. By infusing physics intelligence into artificial intelligence, IAIFI researchers are demonstrating the virtuous cycle of AI+Physics: AI techniques can help make significant discoveries in various physics domains and at the same time physics principles and insights can advance AI tools and lead to improved understanding of AI.

As an illustrative example of Physics impacting AI, IAIFI researchers collaborated across domains to develop a physics-inspired theory of representation learning.² In particular, they improved our understanding of the phenomenon of “grokking,” where models generalize long after overfitting their training set. By analyzing the behavior of the representations in algorithmic tasks, IAIFI researchers were able to explain (and even predict) many aspects of grokking. This work highlights the usefulness of effective theories, commonly used in physics, to model neural network dynamics.

A complementary example, which demonstrates AI impacting Physics, is a project exploring the potential for reinforcement learning to be used to control the squeezed vacuum system of the Laser Interferometer Gravitational-Wave Observatory (LIGO), which would optimize LIGO’s quantum performance.³ LIGO is a flagship NSF experiment with sites in Livingston, Louisiana, and Hanford, Washington, with unparalleled sensitivity to gravitational waves in the world. After an analysis of this technique on past LIGO data, two experimental projects are now being designed to deploy AI on LIGO-like hardware, in simple configurations, and then scale up the complexity to achieve even better sensitivity.

Without dedicated funding to pursue these cross-disciplinary projects, researchers would be limited in their ability to conduct exploratory research that has the potential to significantly impact multiple fields. AI has become pervasive across all fields of science, and a key way to ensure U.S. dominance is to leverage the full spectrum of scientific knowledge to both deploy and strengthen it. Information sharing across disciplines will accelerate advancements in AI by combining domain-specific knowledge and best practices.

²Z. Liu, O. Kitouni, N. Nolte, E.J. Michaud, M. Tegmark and M. Williams, *Towards Understanding Grokking: An Effective Theory of Representation Learning in Conference on Neural Information Processing Systems*, 2022

³C. Whittle, G. Yang, M. Evans and L. Barsotti, Machine learning for quantum-enhanced gravitational-wave observatories, *Physical Review D* 108 (2023)

The RFI notes that a goal of the Administration is that “unnecessarily burdensome requirements do not hamper private sector AI innovation.” One way to achieve this goal is to prioritize robust and interpretable AI from the ground up. As discussed above, Physics has shown to be especially useful as both a testbed for robust AI, and for thinking about AI techniques in a way that can help understand them. Leveraging fundamental science domains like Physics will help develop a deeper understanding of AI through scientific principles.

We recommend that **the Administration direct funding agencies, such as the NSF and DOE, to prioritize supporting interdisciplinary AI+Science programs that focus both on developing AI for scientific discovery and leveraging scientific expertise for AI research.** Existing programs like the NSF AI Institutes enable interdisciplinary collaboration and demonstrate the virtuous cycle of AI and Science.

Educate and train an AI-enabled STEM workforce

The future of AI research and the United States’ strong role in its development is contingent on educating and training an AI-enabled STEM workforce. Regardless of their field, it is becoming increasingly necessary that early-career researchers incorporate AI and related techniques involving data and computer science into their work. Federal support for educating researchers across disciplines and at all education and career stages will help to ensure a pipeline of AI literacy and establish a baseline of knowledge to build upon.

As an example, the IAIFI Fellowship program, which employs postdoctoral researchers working at the intersection of AI and Physics, has demonstrated the impact that dedicated interdisciplinary training can have in both academia and industry. The first cohort of Fellows have moved on to professor positions at leading universities and research positions at top AI companies, such as Meta and Anthropic. AI education at all levels is advantageous for the United States workforce broadly.

In addition, IAIFI runs a week-long graduate-level training program each summer offering opportunities for students to strengthen their skills through lectures and hands-on training. The number of applicants to this program has increased significantly each year, to over 400 applicants in 2025. IAIFI has also worked with MIT to develop an interdisciplinary PhD in Physics, Statistics, and Data Science, and to bolster AI-related curriculum offerings in the Physics department. This includes the development of a free course on the MITx digital learning platform on Computational Data Science in Physics, which has attracted students from around the United States. The enthusiasm for these resources and programs demonstrates the need for this type of interdisciplinary training, especially at the university level.

Many federal funding agencies, *e.g.* the NSF, already consider broader societal impacts in funding decisions. We recommend that **the Administration direct funding agencies to consider AI education, and training an AI-enabled STEM workforce, as high-priority areas** of broader societal impact when evaluating grant proposals.

Provide national infrastructure for scalable computing resources

The use of AI across all stages of research means that there is a broad spectrum of computing needs—some researchers can compute what they need to on their laptops, some need access to several GPUs, and some need sustained access to a much higher volume of resources. The federal government has already begun to set up opportunities for U.S. researchers to access free computing resources through programs such as the National AI Research Resource (NAIRR) Pilot⁴ and the NSF ACCESS program.⁵ The biggest impact of these programs is that they allow researchers at all career stages to access compute, regardless of what their university or company is able to provide.

IAIFI researchers have had the opportunity to use these resources and they are extremely valuable, particularly at the scale of several GPUs for a small project. However, these resources provide little flexibility for scaling up as projects build on the successful development and deployment of AI techniques. Providing scalable resources at the national level would significantly boost the ability of U.S. researchers to conduct meaningful research with AI and would improve the efficiency with which they use the resources.

In order to provide accessible and scalable computing resources, the Administration should build on the NAIRR and ACCESS programs to include options for research groups to request a pool of resources to be allocated to the group for exploratory use, which can then grow as they grow, with the goal of facilitating efficient large-scale deployment of cutting-edge AI technologies.

Conclusion

IAIFI has seen firsthand the impact of AI, not only on our research but also on our early-career talent and our interdisciplinary community. Now is the time to establish how the AI research ecosystem of the United States should operate. We believe that by enabling interdisciplinary research that prioritizes robust and interpretable AI, building an AI-enabled STEM workforce, and providing access to scalable computing resources, **the United States will continue to lead the way in AI globally**, as well as increase the longevity of AI's positive impact on society.

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⁴<https://nairrpilot.org>

⁵<https://access-ci.org>